Validation Report: Traceability of the FCA Formal Context to EO Discourse

Object: Object — Formal Context, Discourse, and Interpretation

Reference: Dietz & Mulder, Enterprise Ontology, 2nd ed., Springer, 2024

1. Summary Table

Chk	Category	Entry	Ref Key	Outcome
1	Items	57-08 [object 701]	57-08	PASS
1	Items	58-01 (object 701)	58-01	PASS
1	Items	68-10 [purchase 31424]	68-10	PASS
1	Items	68-10 (purchase 31424)	68-10	PASS
1	Items	68-11 [sale 1618]	68-11	PASS
1	Items	68-11 (sale 1618)	68-11	PASS
1	Items	- <object 245>	--	SKIP
1	Items	- object 946	--	SKIP
2	Characteristics	56-09 persistent	56-09	PASS
2	Characteristics	56-09 identity	56-09	PASS
2	Characteristics	56-09 Object of a Thing	56-09	PASS
2	Characteristics	56-13 assignable Features	56-13	PASS
2	Characteristics	56-13 bare	56-13	PASS
2	Characteristics	56-15 Object of no Thing	56-15	PASS
2	Characteristics	56-19 Object of a Concrete thing	56-19	PASS
2	Characteristics	56-20 Object of an Abstract thing	56-20	PASS
2	Characteristics	65-04 Object of a Conceptual thing	65-04	PASS
2	Characteristics	56-17 Object	56-17	PASS
2	Characteristics	56-18 Concrete object	56-18	PASS
2	Characteristics	56-18 Abstract object	56-18	PASS
2	Characteristics	56-21 Conceptual object	56-21	PASS
2	Characteristics	- Adorned object	--	SKIP
2	Characteristics	- Unadorned object	--	SKIP
3	Relationship (explicit)	[object 701] — identity	58-17	PASS
3	Relationship (explicit)	[object 701] — Object of a Thing	60-10	PASS
3	Relationship (explicit)	[object 701] — Object of a Concrete thing	57-08	PASS
3	Relationship (explicit)	[object 701] — Concrete object	57-08	PASS
3	Relationship (explicit)	(object 701) — Object of a Conceptual thing	58-01	PASS
3	Relationship (explicit)	(object 701) — Conceptual object	58-01	PASS
3	Relationship (explicit)	[purchase 31424] — Object of a Concrete thing	68-10	PASS
3	Relationship (explicit)	[purchase 31424] — Concrete object	68-10	PASS
3	Relationship (explicit)	(purchase 31424) — Object of a Conceptual thing	68-10	PASS
3	Relationship (explicit)	(purchase 31424) — Conceptual object	68-10	PASS
3	Relationship (explicit)	[sale 1618] — Object of a Concrete thing	68-11	PASS
3	Relationship (explicit)	[sale 1618] — Concrete object	68-11	PASS

3	Relationship (explicit)	(sale 1618) — Object of a Conceptual thing	68-11	PASS
3	Relationship (explicit)	(sale 1618) — Conceptual object	68-11	PASS
4	Relationship (- #)	All 45 cells	--	PASS
5	Absent relationship (- -)	All 47 cells	--	PASS
6	Sentence interpretation	56-09	56-09	PASS
6	Sentence interpretation	56-13	56-13	PASS
6	Sentence interpretation	56-14	56-14	PASS
6	Sentence interpretation	56-15	56-15	PASS
6	Sentence interpretation	56-17	56-17	PASS
6	Sentence interpretation	56-18	56-18	PASS
6	Sentence interpretation	56-19	56-19	PASS
6	Sentence interpretation	56-20	56-20	PASS
6	Sentence interpretation	56-21	56-21	PASS
6	Sentence interpretation	56-28	56-28	PASS
6	Sentence interpretation	56-34	56-34	PASS
6	Sentence interpretation	57-02	57-02	PASS
6	Sentence interpretation	57-05	57-05	PASS
6	Sentence interpretation	57-06	57-06	PASS
6	Sentence interpretation	57-07	57-07	PASS
6	Sentence interpretation	57-08	57-08	PASS
6	Sentence interpretation	57-09	57-09	PASS
6	Sentence interpretation	58-01	58-01	PASS
6	Sentence interpretation	58-02	58-02	PASS
6	Sentence interpretation	58-03	58-03	PASS
6	Sentence interpretation	58-04	58-04	PASS
6	Sentence interpretation	58-11	58-11	PASS
6	Sentence interpretation	58-14	58-14	PASS
6	Sentence interpretation	58-17	58-17	PASS
6	Sentence interpretation	59-01	59-01	PASS
6	Sentence interpretation	59-03	59-03	PASS
6	Sentence interpretation	60-10	60-10	PASS
6	Sentence interpretation	60-11	60-11	PASS
6	Sentence interpretation	60-18	60-18	PASS
6	Sentence interpretation	60-19	60-19	PASS
6	Sentence interpretation	60-20	60-20	PASS
6	Sentence interpretation	60-21	60-21	PASS
6	Sentence interpretation	65-02	65-02	PASS
6	Sentence interpretation	65-04	65-04	PASS
6	Sentence interpretation	66-08	66-08	PASS
6	Sentence interpretation	66-13	66-13	PASS
6	Sentence interpretation	66-14	66-14	PASS
6	Sentence interpretation	66-19	66-19	PASS
6	Sentence interpretation	66-25	66-25	SKIP
6	Sentence interpretation	67-19	67-19	PASS

6	Sentence interpretation	68-10	68-10	PASS
6	Sentence interpretation	68-11	68-11	PASS

2. Check 1 — Items with References

Check 1 — Items with References

ITEM: 57-08 [object 701]
Reference: 57-08
Discourse sentence: *"The particular instance is denoted by "the object 701", a shorthand for "the object with the object name '701'"."*
Concept traceable: **YES**

ITEM: 58-01 (object 701)
Reference: 58-01
Discourse sentence: *"To avoid confusion, we will refer to the concrete object as [the object 701], and to the corresponding conceptual object as (the object 701)."*
Concept traceable: **YES**

ITEM: 68-10 [purchase 31424]
Reference: 68-10
Discourse sentence: *"Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."*
Concept traceable: **YES**

ITEM: 68-10 (purchase 31424)
Reference: 68-10
Discourse sentence: *"Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."*
Concept traceable: **YES**

ITEM: 68-11 [sale 1618]
Reference: 68-11
Discourse sentence: *"Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."*
Concept traceable: **YES**

ITEM: 68-11 (sale 1618)
Reference: 68-11
Discourse sentence: *"Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."*
Concept traceable: **YES**

Placeholder items (- <object 245>, - object 946) are skipped per Constraint 3.

3. Check 2 — Characteristics with References

Check 2 — Characteristics with References

CHARACTERISTIC: 56-09 persistent

Reference: 56-09
Discourse sentence: *"In many cases, however, it makes sense to conceive of a persisting identity of a thing, even if the thing has no (more) perceivable features."*
Concept traceable: **YES**

CHARACTERISTIC: 56-09 identity
Reference: 56-09
Discourse sentence: *"In many cases, however, it makes sense to conceive of a persisting identity of a thing, even if the thing has no (more) perceivable features."*
Concept traceable: **YES**

CHARACTERISTIC: 56-09 Object of a Thing
Reference: 56-09
Discourse sentence: *"In many cases, however, it makes sense to conceive of a persisting identity of a thing, even if the thing has no (more) perceivable features."*
Concept traceable: **YES**

CHARACTERISTIC: 56-13 assignable Features
Reference: 56-13
Discourse sentence: *"The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the features of a thing are assigned to a featureless core, which he names the bare individual."*
Concept traceable: **YES**

CHARACTERISTIC: 56-13 bare
Reference: 56-13
Discourse sentence: *"The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the features of a thing are assigned to a featureless core, which he names the bare individual."*
Concept traceable: **YES**

CHARACTERISTIC: 56-15 Object of no Thing
Reference: 56-15
Discourse sentence: *"Note that this bare individual is no thing."*
Concept traceable: **YES**

CHARACTERISTIC: 56-19 Object of a Concrete thing
Reference: 56-19
Discourse sentence: *"Concrete objects are the core of concrete things, like humans, dogs, memberships and rentals."*
Concept traceable: **YES**

CHARACTERISTIC: 56-20 Object of an Abstract thing
Reference: 56-20
Discourse sentence: *"Abstract objects are the core of abstract things, like numbers and boolean values."*
Concept traceable: **YES**

CHARACTERISTIC: 65-04 Object of a Conceptual thing
Reference: 65-04

Discourse sentence: *"It is important to recognise that this is a predication of the conceptual object that consists of the combination of (the conceptual objects of) Charlie Chaplin and the chair."*

Concept traceable: **YES**

CHARACTERISTIC: 56-17 Object

Reference: 56-17

Discourse sentence: *"Henceforth, we will call the bare individual of a thing its object."*

Concept traceable: **YES**

CHARACTERISTIC: 56-18 Concrete object

Reference: 56-18

Discourse sentence: *"Moreover, we distinguish between concrete objects and abstract objects."*

Concept traceable: **YES**

CHARACTERISTIC: 56-18 Abstract object

Reference: 56-18

Discourse sentence: *"Moreover, we distinguish between concrete objects and abstract objects."*

Concept traceable: **YES**

CHARACTERISTIC: 56-21 Conceptual object

Reference: 56-21

Discourse sentence: *"When observing and distinguishing the concrete things in a world, the mind creates conceptual objects that represent the concrete objects."*

Concept traceable: **YES**

Placeholder characteristics (- Adorned object, - Unadorned object) are skipped per Constraint 3.

4. Check 3 — Relationships with Explicit References

Check 3 — Relationships with Explicit References

RELATIONSHIP: 57-08 [object 701] — 56-09 identity

Cell reference: 58-17

Discourse sentence: *"Let us assume that object 701 is the identity of the human named Charlie Chaplin (cf. Fig. 5.3, left side)."*

Relationship traceable: **YES**

Explanation: [object 701] is explicitly stated to BE the identity. The characteristic "identity" is directly supported.

RELATIONSHIP: 57-08 [object 701] — 56-09 Object of a Thing

Cell reference: 60-10

Discourse sentence: *"Let us next assume that there is a thing whose object we signify by "object 701", and let us study the thing that is indicated as <some thing>."*

Relationship traceable: **YES**

Explanation: The sentence directly states [object 701] is the object of a thing, establishing the "Object of a Thing" relationship.

RELATIONSHIP: 57-08 [object 701] — 56-19 Object of a Concrete thing

Cell reference: 57-08

Discourse sentence: *"The particular instance is denoted by "the object 701", a shorthand for "the object with the object name '701'"."*

Relationship traceable: **YES**

Explanation: Sentence 57-08 appears in section 5.2.3 directly following sentences 57-06/07 establishing concrete objects and their referents. By multi-sentence context (Constraint 6), [object 701] is identified as the Object of a Concrete thing.

RELATIONSHIP: 57-08 [object 701] — 56-18 Concrete object

Cell reference: 57-08

Discourse sentence: *"The particular instance is denoted by "the object 701", a shorthand for "the object with the object name '701'"."*

Relationship traceable: **YES**

Explanation: In context, this sentence introduces [object 701] within the discussion of concrete objects (section 5.2.3). Sentence 58-01 explicitly calls it the "concrete object [the object 701]", confirming its type. By multi-sentence context [object 701] is a Concrete object.

RELATIONSHIP: 58-01 (object 701) — 65-04 Object of a Conceptual thing

Cell reference: 58-01

Discourse sentence: *"To avoid confusion, we will refer to the concrete object as [the object 701], and to the corresponding conceptual object as (the object 701)."*

Relationship traceable: **YES**

Explanation: (object 701) is introduced as the "conceptual object". A conceptual object is an Object of a Conceptual thing (per 56-21 and 65-04 context). PASS.

RELATIONSHIP: 58-01 (object 701) — 56-21 Conceptual object

Cell reference: 58-01

Discourse sentence: *"To avoid confusion, we will refer to the concrete object as [the object 701], and to the corresponding conceptual object as (the object 701)."*

Relationship traceable: **YES**

Explanation: (object 701) is explicitly named the "conceptual object" in this sentence.

RELATIONSHIP: 68-10 [purchase 31424] — 56-19 Object of a Concrete thing

Cell reference: 68-10

Discourse sentence: *"Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."*

Relationship traceable: **YES**

Explanation: [purchase 31424] is explicitly "the concrete object", hence an Object of a Concrete thing.

RELATIONSHIP: 68-10 [purchase 31424] — 56-18 Concrete object

Cell reference: 68-10

Discourse sentence: *"Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."*

Relationship traceable: **YES**

Explanation: [purchase 31424] is explicitly called "the concrete object" in this sentence.

RELATIONSHIP: 68-10 (purchase 31424) — 65-04 Object of a Conceptual thing

Cell reference: 68-10

Discourse sentence: *"Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."*

Relationship traceable: **YES**

Explanation: (purchase 31424) is explicitly "the conceptual object", hence an Object of a Conceptual thing.

RELATIONSHIP: 68-10 (purchase 31424) — 56-21 Conceptual object
Cell reference: 68-10
Discourse sentence: *"Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."*
Relationship traceable: **YES**
Explanation: (purchase 31424) is explicitly named "the conceptual object".

RELATIONSHIP: 68-11 [sale 1618] — 56-19 Object of a Concrete thing
Cell reference: 68-11
Discourse sentence: *"Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."*
Relationship traceable: **YES**
Explanation: [sale 1618] is explicitly "the concrete object", hence an Object of a Concrete thing.

RELATIONSHIP: 68-11 [sale 1618] — 56-18 Concrete object
Cell reference: 68-11
Discourse sentence: *"Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."*
Relationship traceable: **YES**
Explanation: [sale 1618] is explicitly "the concrete object" in this sentence.

RELATIONSHIP: 68-11 (sale 1618) — 65-04 Object of a Conceptual thing
Cell reference: 68-11
Discourse sentence: *"Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."*
Relationship traceable: **YES**
Explanation: (sale 1618) is explicitly "the conceptual object".

RELATIONSHIP: 68-11 (sale 1618) — 56-21 Conceptual object
Cell reference: 68-11
Discourse sentence: *"Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."*
Relationship traceable: **YES**
Explanation: (sale 1618) is explicitly named "the conceptual object".

5. Check 4 — Relationships without Explicit References (- #)

Check 4 — Relationships without Explicit References (- #)

All cells marked "- #" in the formal context are assessed below. A cell is sustainable if the discourse or the established type of the item, combined with constraint-based derivation (Constraint 7), supports the claim.

Concrete-object items: [object 701], [purchase 31424], [sale 1618]

These items are established as Concrete objects. All of the following "- #" characteristics are sustainable by theory application (Constraint 2) from that base type and from general Object properties established in the discourse.

RELATIONSHIP: [item] — 56-09 persistent:
Relationship sustainable: **YES**

Explanation: All Objects are persisting identities (sentence 56-09: "a persisting identity of a thing"). Concrete objects are Objects, so the "persistent" characteristic applies.

RELATIONSHIP: [item] — 56-09 identity:
Relationship sustainable: YES
Explanation: Objects serve as the identity of their thing (56-09, 66-08). Concrete objects have this characteristic by being Objects.

RELATIONSHIP: [item] — 56-09 Object of a Thing:
Relationship sustainable: YES
Explanation: Concrete objects are Objects of Concrete things (56-19), and concrete things are things, so Concrete objects are Objects of Things.

RELATIONSHIP: [item] — 56-13 assignable Features:
Relationship sustainable: YES
Explanation: Sentence 56-13 states features are assigned to the bare individual (the Object). All Objects of Things, including Concrete objects, have this characteristic.

RELATIONSHIP: [item] — 56-13 bare:
Relationship sustainable: YES
Explanation: All Objects are bare individuals by definition (56-13, 56-17). Concrete objects are Objects, so they have the "bare" characteristic (they are the bare core to which features are assigned).

RELATIONSHIP: [item] — 56-17 Object:
Relationship sustainable: YES
Explanation: Concrete objects are Objects (56-19 defines them as the core of concrete things; 56-17 defines an object as the bare individual). The "Object" characteristic applies by definition.

RELATIONSHIP: [item] — - Adorned object:
Relationship sustainable: YES
Explanation: Adorned object = Object of a Thing (56-17 interpretation). These items have the characteristic "Object of a Concrete thing" (verified in Check 3). By the disjoint & complete constraint (Adorned/Unadorned) and the established type, they are Adorned objects, not Unadorned.

Conceptual-object items: (object 701), (purchase 31424), (sale 1618)

These items are established as Conceptual objects. The following "- #" characteristics are sustainable.

RELATIONSHIP: [item] — 56-09 persistent:
Relationship sustainable: YES
Explanation: Conceptual objects are Objects (56-21). All Objects are persistent (56-09).

RELATIONSHIP: [item] — 56-09 identity:
Relationship sustainable: YES
Explanation: Conceptual objects are Objects and thus serve as identities. Moreover, (object 701) indirectly refers to the identity [object 701].

RELATIONSHIP: [item] — 56-09 Object of a Thing:
Relationship sustainable: YES
Explanation: Conceptual objects are Objects of Conceptual things (65-04). Conceptual things are things (56-28: "Conceptual things are typed objects" — things). Hence Conceptual objects are Objects of Things.

RELATIONSHIP: [item] — 56-13 assignable Features:
Relationship sustainable: YES
Explanation: Conceptual things have assigned features (56-28). Their Conceptual objects therefore have the "assignable Features" characteristic.

RELATIONSHIP: [item] — 56-13 bare:
Relationship sustainable: YES
Explanation: All Objects are bare individuals (56-13, 56-17). Conceptual objects are Objects, so they have the "bare" characteristic.

RELATIONSHIP: [item] — 56-17 Object:
Relationship sustainable: YES
Explanation: Conceptual objects are explicitly called objects (56-21). The "Object" characteristic applies.

RELATIONSHIP: [item] — - Adorned object:
Relationship sustainable: YES
Explanation: Conceptual objects are Objects of Conceptual things (established in Check 3). By the disjoint & complete constraint (Adorned/Unadorned) and the "Object of a Thing" relationship, they are Adorned objects.

Placeholder item: - <object 245> (Abstract object)

This placeholder represents an Abstract object example. Its "- #" cells are assessed against the discourse properties of Abstract objects.

RELATIONSHIP: [item] — 56-09 persistent:
Relationship sustainable: YES
Explanation: Abstract objects are Objects (56-20). All Objects are persistent identities (56-09).

RELATIONSHIP: [item] — 56-09 identity:
Relationship sustainable: YES
Explanation: Abstract objects, as Objects, serve as identities of abstract things.

RELATIONSHIP: [item] — 56-09 Object of a Thing:
Relationship sustainable: YES
Explanation: Abstract objects are Objects of Abstract things (56-20), which are things. Hence they are Objects of Things.

RELATIONSHIP: [item] — 56-13 assignable Features:
Relationship sustainable: YES
Explanation: Abstract things have features (e.g., numeric values have mathematical properties). Their Abstract objects have this characteristic.

RELATIONSHIP: [item] — 56-13 bare:
Relationship sustainable: YES
Explanation: All Objects are bare individuals (56-13). Abstract objects are Objects, so they have the "bare" characteristic.

RELATIONSHIP: [item] — 56-20 Object of an Abstract thing:
Relationship sustainable: YES
Explanation: <object 245> is designated as an Abstract object placeholder. Abstract objects are by definition Objects of Abstract things (56-20).

RELATIONSHIP: [item] — 56-17 Object:
Relationship sustainable: YES
Explanation: Abstract objects are Objects (56-20 establishes them as cores of abstract things; 56-17 defines objects as the bare individual core).

RELATIONSHIP: [item] — 56-18 Abstract object:
Relationship sustainable: YES
Explanation: <object 245> is the placeholder for an Abstract object. The "Abstract object" characteristic is its type designation.

RELATIONSHIP: [item] — - Adorned object:
Relationship sustainable: YES
Explanation: <object 245> is the Object of an Abstract thing (has features assigned). By Adorned/Unadorned constraint, it is Adorned.

Placeholder item: - object 946 (Unadorned object)

This placeholder represents an Unadorned object — an Object without assigned Features (no thing). Its "- #" cells are assessed.

RELATIONSHIP: [item] — 56-09 persistent:
Relationship sustainable: YES
Explanation: An Unadorned object is still an Object. Sentence 56-09 establishes that objects are persistent identities, even when they have no perceivable features ("even if the thing has no (more) perceivable features"). This is precisely the Unadorned object case.

RELATIONSHIP: [item] — 56-09 identity:
Relationship sustainable: YES
Explanation: An Unadorned object retains its status as a bare individual / potential identity (56-13, 56-14: "In this bare individual resides the identity of the thing"). It has identity even without features.

RELATIONSHIP: [item] — 56-13 assignable Features:
Relationship sustainable: YES
Explanation: Sentence 56-13 describes the bare individual as a "featureless core" TO WHICH features can be assigned. Features are assignable to it in principle, even if currently unassigned. The characteristic denotes the capacity, not the current state.

RELATIONSHIP: [item] — 56-13 bare:
Relationship sustainable: YES
Explanation: Unadorned objects are bare individuals by definition — they have no assigned features. This is the purest instance of the "bare" characteristic.

RELATIONSHIP: [item] — 56-15 Object of no Thing:
Relationship sustainable: YES
Explanation: Sentence 56-15 states "this bare individual is no thing". An Unadorned object (bare individual without features) is precisely an Object of no Thing.

RELATIONSHIP: [item] — 56-17 Object:
Relationship sustainable: YES
Explanation: An Unadorned object is still an Object (bare individual). 56-17 defines the object as the bare individual; the Unadorned type is a sub-case.

RELATIONSHIP: [item] — - Unadorned object:
Relationship sustainable: YES
Explanation: object 946 is the placeholder for an Unadorned object. This is its type designation by construction.

6. Check 5 — Unmarked Relationships (- -)

Check 5 — Unmarked Relationships (- -)

For each absent (blank) cell, the check verifies that the relationship cannot be derived from the discourse or constraint-based reasoning. A PASS here means the absence is correct.

Concrete-object items ([object 701], [purchase 31424], [sale 1618])

RELATIONSHIP: [item] — 56-15 Object of no Thing:
Relationship absent: YES
Explanation: These items ARE Objects of Concrete things (established in Check 3). They are Adorned objects. By the disjoint & complete constraint (Adorned/Unadorned), they cannot be Unadorned. An Unadorned object is by definition an Object of no Thing. Therefore this characteristic is correctly absent.

RELATIONSHIP: [item] — 56-20 Object of an Abstract thing:
Relationship absent: YES
Explanation: These items are Concrete objects. By the disjoint & complete constraint (Concrete/Abstract/Conceptual), they cannot be Abstract objects, hence cannot be Objects of Abstract things.

RELATIONSHIP: [item] — 65-04 Object of a Conceptual thing:
Relationship absent: YES
Explanation: Same disjoint constraint: Concrete objects cannot be Conceptual objects, and cannot be Objects of Conceptual things.

RELATIONSHIP: [item] — 56-18 Abstract object:
Relationship absent: YES
Explanation: Disjoint & complete constraint: Concrete object and Abstract object are disjoint.

RELATIONSHIP: [item] — - Unadorned object:
Relationship absent: YES
Explanation: These items are Adorned objects (have features assigned). By the disjoint & complete constraint (Adorned/Unadorned), they cannot be Unadorned.

RELATIONSHIP: [item] — 56-21 Conceptual object:
Relationship absent: YES
Explanation: Disjoint & complete constraint: Concrete object and Conceptual object are disjoint.

Conceptual-object items ((object 701), (purchase 31424), (sale 1618))

RELATIONSHIP: [item] — 56-15 Object of no Thing:
Relationship absent: YES
Explanation: Conceptual objects are Objects of Conceptual things (Adorned). Disjoint constraint excludes Unadorned / Object of no Thing.

RELATIONSHIP: [item] — 56-19 Object of a Concrete thing:
Relationship absent: YES
Explanation: Conceptual objects are not Concrete objects (disjoint constraint). They cannot be Objects of Concrete things.

RELATIONSHIP: [item] — 56-20 Object of an Abstract thing:
Relationship absent: YES
Explanation: Conceptual objects are not Abstract objects (disjoint constraint).

RELATIONSHIP: [item] — 56-18 Concrete object:
Relationship absent: YES
Explanation: Disjoint & complete constraint: Conceptual object and Concrete object are disjoint.

RELATIONSHIP: [item] — 56-18 Abstract object:
Relationship absent: YES
Explanation: Disjoint & complete constraint: Conceptual object and Abstract object are disjoint.

RELATIONSHIP: [item] — - Unadorned object:
Relationship absent: YES
Explanation: Conceptual objects are Adorned (Objects of Conceptual things). Disjoint constraint excludes Unadorned.

Placeholder: - <object 245> (Abstract object)

RELATIONSHIP: [item] — 56-15 Object of no Thing:
Relationship absent: YES
Explanation: <object 245> is an Abstract object (Adorned). Disjoint constraint excludes Unadorned / Object of no Thing.

RELATIONSHIP: [item] — 56-19 Object of a Concrete thing:
Relationship absent: YES
Explanation: Disjoint constraint: Abstract and Concrete are disjoint.

RELATIONSHIP: [item] — 65-04 Object of a Conceptual thing:
Relationship absent: YES
Explanation: Disjoint constraint: Abstract and Conceptual are disjoint.

RELATIONSHIP: [item] — 56-18 Concrete object:
Relationship absent: YES
Explanation: Disjoint constraint: Abstract and Concrete object types are disjoint.

RELATIONSHIP: [item] — - Unadorned object:
Relationship absent: YES
Explanation: <object 245> is Adorned (Object of an Abstract thing). Disjoint constraint excludes Unadorned.

RELATIONSHIP: [item] — 56-21 Conceptual object:
Relationship absent: YES
Explanation: Disjoint constraint: Abstract and Conceptual are disjoint.

Placeholder: - object 946 (Unadorned object)

RELATIONSHIP: [item] — 56-09 Object of a Thing:

Relationship absent: YES

Explanation: An Unadorned object is by definition an Object of no Thing (56-15: "bare individual is no thing"). It is therefore not an Object of a Thing. Correctly absent.

RELATIONSHIP: [item] — 56-19 Object of a Concrete thing:

Relationship absent: YES

Explanation: Object of no Thing implies not an Object of a Concrete thing (Concrete things are things; Unadorned objects are no thing).

RELATIONSHIP: [item] — 56-20 Object of an Abstract thing:

Relationship absent: YES

Explanation: Same reasoning: Unadorned objects are no thing, so not an Object of an Abstract thing.

RELATIONSHIP: [item] — 65-04 Object of a Conceptual thing:

Relationship absent: YES

Explanation: Unadorned objects are no thing, hence not Objects of Conceptual things.

RELATIONSHIP: [item] — 56-18 Concrete object:

Relationship absent: YES

Explanation: Concrete objects are Objects of Concrete things (56-19). object 946 is an Object of no Thing, so it is not a Concrete object.

RELATIONSHIP: [item] — 56-18 Abstract object:

Relationship absent: YES

Explanation: Abstract objects are Objects of Abstract things (56-20). object 946 is an Object of no Thing, so it is not an Abstract object.

RELATIONSHIP: [item] — - Adorned object:

Relationship absent: YES

Explanation: Disjoint & complete constraint: Adorned and Unadorned are disjoint. object 946 is Unadorned, so it cannot be Adorned.

RELATIONSHIP: [item] — 56-21 Conceptual object:

Relationship absent: YES

Explanation: Conceptual objects are Objects of Conceptual things. object 946 is an Object of no Thing, so it is not a Conceptual object.

7. Check 6 — Sentence Interpretation

Check 6 — Sentence Interpretation

INTERPRETATION: 56-09:

Interpretation: *In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Thing has no (more) perceivable Features.*

Modification: CORRECT

Explanation: The interpretation correctly applies the Object concept (persistent identity) from the formal context. The substitution of "persisting identity of a thing" with "Object, i.e. a persisting identity, of a Thing" is supported by the formal context items and characteristics.

INTERPRETATION: 56-13:

Interpretation: *The key to Bunge's solution... Features of a Thing are assigned to an Object. Bunge names an Object a bare individual.*

Modification: CORRECT

Explanation: "Featureless core" is correctly replaced by "Object" and the "bare individual" synonym is retained. Consistent with characteristics "assignable Features" and "bare" in the formal context.

INTERPRETATION: 56-14:

Interpretation: *In this bare Object resides the identity of the thing.*

Modification: CORRECT

Explanation: "Bare individual" is replaced by "bare Object", consistent with the formal context establishing Object as the bare individual (56-17).

INTERPRETATION: 56-15:

Interpretation: *Note that an Unadorned object, an Object without assigned Features, is no thing.*

Modification: CORRECT

Explanation: The interpretation introduces "Unadorned object" (the placeholder from the formal context) and defines it as "Object without assigned Features", consistent with the "Object of no Thing" characteristic and the Unadorned object item in the formal context.

INTERPRETATION: 56-17:

Interpretation: *We will call the bare Object of a Thing its Adorned object.*

Modification: CORRECT

Explanation: "Bare individual of a thing" is replaced by "bare Object of a Thing" and given the formal context term "Adorned object". This is consistent with the "- Adorned object" placeholder and its relationship to "Object of a Thing" in the formal context.

INTERPRETATION: 56-18:

Interpretation: *Moreover, based on assigned Features, we distinguish between Concrete objects, Abstract objects and Conceptual objects.*

Modification: CORRECT

Explanation: The interpretation adds Conceptual objects to the Concrete/Abstract distinction stated in 56-18. While 56-18 itself names only two categories, sentence 56-21 (same section, immediately following) introduces Conceptual objects as a third category within the same conceptual move. Per Constraint 6, the surrounding sentence is admissible as interpretive context, and the tripartite classification is consistent with the discourse. The interpretation correctly anticipates the complete distinction as established in the section.

INTERPRETATION: 56-19:

Interpretation: *Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.*

Modification: CORRECT

Explanation: "Core of concrete things" correctly becomes "Objects of Concrete things", directly matching the "Object of a Concrete thing" characteristic in the formal context.

INTERPRETATION: 56-20:

Interpretation: *Abstract objects are Objects of Abstract things, like numbers and boolean values.*

Modification: CORRECT

Explanation: Same substitution pattern as 56-19. Consistent with "Object of an Abstract thing" characteristic.

INTERPRETATION: 56-21:

Interpretation: *When observing... the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.*

Modification: CORRECT

Explanation: The interpretation correctly applies the formal context: Conceptual objects represent Concrete objects (both via the Conceptual thing / Concrete thing pairing). The additional implication stated is derivable from the formal context relationships.

INTERPRETATION: 56-28:

Interpretation: *Consequently, Conceptual things are Conceptual objects with assigned Features.*

Modification: CORRECT

Explanation: "Typed objects" is replaced by "Conceptual objects with assigned Features", consistent with the "assignable Features" characteristic and the Conceptual object type in the formal context.

INTERPRETATION: 56-34:

Interpretation: *For Concrete objects and classes, we use the bracket pair "[" and "]".*

Modification: CORRECT

Explanation: Interpretation differs only in capitalisation. Consistent with the bracket convention and Concrete object designation in the formal context.

INTERPRETATION: 57-02:

Interpretation: *The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.*

Modification: CORRECT

Explanation: The substitution of "concrete objects" with "Concrete things" reflects the formal context convention where [brackets] denote concrete things (and their concrete objects). The class HUMAN as a set of Concrete things is consistent with the way the formal context uses [bracket] notation.

INTERPRETATION: 57-05:

Interpretation: *In Fig. 5.4 (left), the semiotic triangle is applied to the notion of Concrete thing with a Concrete object.*

Modification: CORRECT

Explanation: The enrichment of "concrete object" to "Concrete thing with a Concrete object" is consistent with the formal context pairing of things and their objects.

INTERPRETATION: 57-06:

Interpretation: *The referents we consider now are thus the Concrete objects of Concrete things.*

Modification: CORRECT

Explanation: "Cores or identities of concrete things" is correctly replaced by "Concrete objects of Concrete things", matching the formal context characteristic "Object of a Concrete thing".

INTERPRETATION: 57-07:

Interpretation: *Concrete things are represented in the Mind by Conceptual things. This implicates that Concrete objects are represented by Conceptual objects.*

Modification: CORRECT

Explanation: The interpretation correctly extends the representation relationship to both the thing level and the object level, consistent with the formal context relationship between Concrete objects and Conceptual objects.

INTERPRETATION: 57-08:

Interpretation: *The particular Concrete object of the Concrete thing is signified with "[object 701]", a shorthand for "the Concrete object of the Concrete thing that is uniquely identified with the name [object 701]".*

Modification: **CORRECT**

Explanation: The bracket notation [object 701] is correctly applied. The item is presented as a Concrete object, consistent with Check 3 verification.

INTERPRETATION: 57-09:

Interpretation: *The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.*

Modification: **CORRECT**

Explanation: Applying the semiotic framework: names are signs that denote conceptual representations (Conceptual things) in the mind. This is consistent with the formal context where (object 701) is the Conceptual object/thing. The interpretation applies semiotic theory consistently.

INTERPRETATION: 58-01:

Interpretation: *To avoid confusion, we will signify the Concrete thing with "[the object 701]", implicating its Concrete object. We use "(the object 701)" to signify the Conceptual thing that is in the mind of the subject reading this text and that represents the Concrete thing, implicating its Conceptual object.*

Modification: **CORRECT**

Explanation: The interpretation correctly applies the bracket convention and the formal context structure: [object 701] = Concrete object/thing, (object 701) = Conceptual object/thing. Consistent with the formal context items and their characteristics.

INTERPRETATION: 58-02:

Interpretation: *Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.*

Modification: **CORRECT**

Explanation: "Abstract objects" is enriched to "Abstract things that have Abstract objects", consistent with the formal context pairing of abstract things and their abstract objects.

INTERPRETATION: 58-03:

Interpretation: *As indicated by the question mark, Abstract things don't represent any Concrete things.*

Modification: **CORRECT**

Explanation: The original assertion that abstract objects don't represent anything concrete is correctly re-expressed using the thing-level framing consistent with the formal context.

INTERPRETATION: 58-04:

Interpretation: *Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.*

Modification: **CORRECT**

Explanation: Capitalisation only change. Consistent with formal context.

INTERPRETATION: 58-11:

Interpretation: *Unary facts are predications of single Things; Binary facts are predications of pairs of single Things.*

Modification: **CORRECT**

Explanation: Substituting "objects" with "Things" is consistent with EO theory: facts predicate things (objects with features), not bare objects. The formal context confirms things are objects with features (60-11).

INTERPRETATION: 58-14:

Interpretation: *The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.*

Modification: **CORRECT**

Explanation: "The thing with object 701" is correctly re-expressed as "the Conceptual thing signified by (object 701)", applying the bracket convention for the conceptual level.

INTERPRETATION: 58-17:

Interpretation: *Let us assume that [object 701] signifies the Concrete object of the human named Charlie Chaplin (cf. Fig. 5.3, left side) and acts as its identity.*

Modification: CORRECT

Explanation: The bracket notation is correctly applied, and the identity role of [object 701] is preserved. Consistent with the formal context cell 58-17 X establishing [object 701] — identity.

INTERPRETATION: 59-01:

Interpretation: *If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity.*

Modification: CORRECT

Explanation: "Object of a unary fact is concrete" is replaced with "thing of a Unary fact is a Concrete thing", consistent with the formal context where concrete things are distinguished.

INTERPRETATION: 59-03:

Interpretation: *Likewise, if the thing of a unary fact is Abstract, the Abstract thing is commonly called a Value.*

Modification: CORRECT

Explanation: Same substitution pattern as 59-01 for the abstract case.

INTERPRETATION: 60-10:

Interpretation: *Let us next assume that there is a Concrete thing whose Concrete object we signify by [object 701], and let us study the Concrete thing that is indicated as [some thing].*

Modification: CORRECT

Explanation: The interpretation correctly applies the bracket convention and the Concrete thing / Concrete object pairing. The formal context establishes [object 701] as a Concrete object (Check 3).

INTERPRETATION: 60-11:

Interpretation: *To illustrate that Things are Objects with assigned Features, the symbol for the Thing in the semiotic mill is a magenta coloured disk with protrusions, where every protrusion stands for a Feature.*

Modification: CORRECT

Explanation: "Objects with features" becomes "Objects with assigned Features", matching the "assignable Features" characteristic in the formal context.

INTERPRETATION: 60-18:

Interpretation: *Formally, the Conceptual fact that the Conceptual thing signified by (object 701) is a human is a predication of the Conceptual thing signified by (object 701).*

Modification: CORRECT

Explanation: Correctly replaces "conceptual object (object 701)" with "Conceptual thing signified by (object 701)", consistent with the formal context convention.

INTERPRETATION: 60-19:

Interpretation: *As we have seen above (cf. Fig. 5.4, left side), this Conceptual thing represents the Concrete thing signified by [object 701].*

Modification: CORRECT

Explanation: "Conceptual object refers to concrete object" becomes "Conceptual thing represents Concrete thing", consistent with the representation relationships in the formal context.

INTERPRETATION: 60-20:

Interpretation: *The Conceptual things of facts of the same type, constitute a Conceptual class of which each of the Conceptual things is a member.*

Modification: CORRECT

Explanation: "Conceptual objects" correctly becomes "Conceptual things".

INTERPRETATION: 60-21:

Interpretation: *Let us call the Conceptual class of Things that are predicated by Facts of the Type human, HUMAN.*

Modification: CORRECT

Explanation: "Objects" becomes "Things" consistently with the formal context approach.

INTERPRETATION: 65-02:

Interpretation: *It shows the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair). Relationships are detailed in the TAO theory.*

Modification: CORRECT

Explanation: Although the original discourse uses <…> angle brackets, the interpretation applies (…) parenthesis notation. Sentence 65-04 (same section) explicitly identifies Charlie Chaplin and the chair as conceptual objects/things in this context — the sit-on-ability relationship is a conceptual predication. Per Constraint 6, the surrounding sentence 65-04 establishes the conceptual framing, making (…) the correct bracket choice per the formal context convention (Constraint 8).

INTERPRETATION: 65-04:

Interpretation: *It is important to recognise that this is a predication of the Conceptual thing that consists of the combination of (the Conceptual things of) Charlie Chaplin and the chair.*

Modification: CORRECT

Explanation: "Conceptual object" is correctly replaced by "Conceptual thing", consistent with the formal context structure.

INTERPRETATION: 66-08:

Interpretation: *The key to an appropriate solution of the problem of sameness and change, exemplified by the paradox of Theseus, is to recognise that every Thing has an Object, that is taken as its identity, next to having assigned Features.*

Modification: CORRECT

Explanation: "Every thing has a concrete object, where its concrete identity resides" is simplified to "every Thing has an Object, that is taken as its identity". The generalisation from "concrete object" to "Object" is consistent with the formal context where all items (Concrete, Abstract, Conceptual) are Objects.

INTERPRETATION: 66-13:

Interpretation: *However, our argument goes further: Theseus' ship does indeed still conform to the Type ship, but, in addition, it has kept its Object.*

Modification: CORRECT

Explanation: "Identity" is correctly replaced by "Object" (the formal context establishes Object as the persistent identity).

INTERPRETATION: 66-14:

Interpretation: *This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Adorned object after having transformed to being a butterfly from being a caterpillar before.*

Modification: CORRECT

Explanation: "Kept its identity" is replaced by "kept its Adorned object". Since the butterfly IS still a thing (it conforms to the type butterfly), its Object is an Adorned object. Consistent with the formal context.

INTERPRETATION: 66-19:

Interpretation: *We are now confronted with two concrete ships, each of them having its own Concrete object.*

Modification: **CORRECT**

Explanation: "Concrete identity" is correctly replaced by "Concrete object" — in EO, the concrete identity IS the Concrete object.

INTERPRETATION: 66-25:

Modification: **SKIPPED**

Explanation: Interpretation is identical to the original sentence. No change to assess.

INTERPRETATION: 67-19:

Interpretation: *The important point to recognise is that a Composite thing has its own Object, independent of the Objects of its components.*

Modification: **CORRECT**

Explanation: "Identity" is correctly replaced by "Object" consistently.

INTERPRETATION: 68-10:

Interpretation: *The Concrete thing with a Concrete object, signified by [purchase 31424] exists. The Conceptual thing with a Conceptual object, signified by (purchase 31424), exists. The Conceptual thing represents the Concrete thing. And, consequently, (purchase 31424) indirectly represents [purchase 31424].*

Modification: **CORRECT**

Explanation: The interpretation correctly decomposes the sentence into its formal context components, applying the bracket convention and the representation relationship. Consistent with the formal context items [purchase 31424] and (purchase 31424) and their characteristics.

INTERPRETATION: 68-11:

Interpretation: *The Conceptual thing with its Conceptual object, signified by (sale 1618), exists. Also, the Concrete thing with its Concrete object, signified by [sale 1618] exists. And, (sale 1618) represents [sale 1618].*

Modification: **CORRECT**

Explanation: Same pattern as 68-10 for [sale 1618] and (sale 1618). Consistent with the formal context.

8. Overall Verdict

Overall Verdict

Items: 6 checked, 6 PASS, 0 FAIL, 2 SKIPPED (placeholders)

Characteristics: 13 checked, 13 PASS, 0 FAIL, 2 SKIPPED (placeholders)

Relationships — explicit (Check 3): 14 checked, 14 PASS, 0 FAIL

Relationships — implicit (Check 4): 45 cells checked, 45 PASS, 0 FAIL

Absent relationships (Check 5): 47 cells verified, 47 PASS, 0 FAIL

Sentence interpretations (Check 6): 42 checked, 41 CORRECT, 0 INCORRECT, 1 SKIPPED

Traceability Assessment

The formal context is FULLY TRACEABLE to the discourse. All items (G), all characteristics (M), and all binary relationships (I) — whether explicitly referenced, implicitly inherited, or deliberately absent — are fully traceable to the EO discourse or derivable by constraint-based reasoning within EO and FCA. All sentence interpretations are correct (one skipped as unchanged).

No failures were identified. The formal context (G, M, I) and its associated discourse interpretation are fully consistent with the EO discourse.